

torch.histogram#

<https://pytorch.org/docs/stable/generated/torch.histogram.html>

Description:

Computes a histogram of the values in a tensor. bins can be an integer or a 1D tensor. If bins is an int, it specifies the number of equal-width bins. By default, the lower and upper range of the bins is determined by the minimum and maximum elements of the input tensor. The range argument can be provided to specify a range for the bins. If bins is a 1D tensor, it specifies the sequence of bin edges including the rightmost edge. It should contain at least 2 elements and its elements should be increasing. input (Tensor) – the input tensor.

Function Signature

```
torch.histogram(input, bins, *, range=None, weight=None, density=False, out=None)#
```

Parameters

Keyword Arguments

Returns

Return type

Parameters

Keyword

range (tuple of float) – Defines the range of the bins. weight (Tensor) – If provided, weight should have the same shape as input. Each value in input contributes its associated weight towards its bin's result. density (bool) – If False, the result will contain the count (or total weight) in each bin. If True, the result is the value of the probability density function over the bins, normalized such that the integral over the range of the bins is 1. out (Tensor, optional) – the output tensor. (tuple, optional): The result tuple of two output tensors (hist, bin_edges).

Returns:

hist (Tensor)

Code Examples

```
>>> torch.histogram(torch.tensor([1., 2, 1]), bins=4, range=(0., 3.), weight=torch.tensor([1., 2., 4.]))
(tensor([ 0.,  5.,  2.,  0.]), tensor([0., 0.75, 1.5, 2.25, 3.]))
>>> torch.histogram(torch.tensor([1., 2, 1]), bins=4, range=(0., 3.), weight=torch.tensor([1., 2., 4.]), density=True)
(tensor([ 0.,  0.9524,  0.3810,  0.]), tensor([0., 0.75, 1.5, 2.25, 3.]))
```

Detailed Documentation

Documentation

Computes a histogram of the values in a tensor.

bins can be an integer or a 1D tensor.

If bins is an int, it specifies the number of equal-width bins. By default, the lower and upper range of the bins is determined by the minimum and maximum elements of the input tensor. The range argument can be provided to specify a range for the bins.

If bins is a 1D tensor, it specifies the sequence of bin edges including the rightmost

edge. It should contain at least 2 elements and its elements should be increasing.

input (Tensor) – the input tensor.

bins – int or 1D Tensor. If int, defines the number of equal-width bins. If tensor, defines the sequence of bin edges including the rightmost edge.

range (tuple of float) – Defines the range of the bins.

weight (Tensor) – If provided, weight should have the same shape as input. Each value in input contributes its associated weight towards its bin's result.

density (bool) – If False, the result will contain the count (or total weight) in each bin. If True, the result is the value of the probability density function over the bins, normalized such that the integral over the range of the bins is 1.

out (Tensor, optional) – the output tensor. (tuple, optional): The result tuple of two output tensors (hist, bin_edges).

1D Tensor containing the values of the histogram. bin_edges(Tensor): 1D Tensor containing the edges of the histogram bins.